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GENETIC ENGINEERING

INTRODUCTION

Genetic engineering (also called genetic modification) is the direct human manipulation of an organism's genome using modern DNA technology. It involves the introduction of foreign DNA or synthetic genes into the organisms of interest.

Popularly referred to as genetic engineering, it came to be defined as the basis or tool of new biotechnology. The term genetic engineering was first coined by Jack Williamson in 1951.

The most common form of genetic engineering involves the insertion of new genetic material at an unspecified location in the host genome. This is accomplished by isolating and copying the genetic material of interest using molecular cloning methods to generate a DNA sequence containing the required genetic elements for expression, and then inserting this constructed into the host organism.

Genetic engineering alters the genetic makeup of an organism using techniques that introduce heritable material prepared outside the organism either directly into the host or into a cell that is then fused or hybridized with the host. This inmoves using recombinant nucleic and (DNA or RNA) techniques to form new combinations of heritable genetic material followed by the incorporation of that material either indirectly through a vector system or directly through micro-injection, macro-injection and micro-encapsulation techniques.

An organism that is generated through the introduction of recombinant DNA is considered to be a genetically modified organism or transgenic organism. The first organisms genetically engineered were – bacteria in 1973 and then mice in 1974. Insulin-producing bacteria were commercialized in 1982, and genetically modified food has been sold since 1994.

All genetic material from another species is added to the host, the resulting organism is called transgenic. If genetic material from the same species or a species that can

naturally breed with the host is used, the resulting organism is called Cisgenic. Genetic engineering can also be used to remove genetic material from the target organism, creating a gene knockout organism.

Applications in Medicine:

Pharmacogenomics it is the study of how the genetic inheritance of an individual affects his/her body's response to drugs. The vision of pharmacogenomics is to be able to design and produce drugs that are adapted to each person's genetic makeup.

Advantages of Pharmacogenomics

1. **Development of tailor-made medicines:** Using pharmacogenomics, pharmaceutical companies can create drugs based on proteins, enzymes and RNA molecules that are associated with specific genes and diseases. These tailor-made drugs promise not only to maximize therapeutic effects but also to degrease damaged to nearby healthy cells.
2. **More accurate methods of determining appropriate drug dosages:** Knowing a patient's genetics will enable doctors to determine how well his/her body can process and metabolize a medicine. This will maximize the value of medicine and decrease the likelihood of overdose.
3. **Improvements in the drug discovery and approval process:** Genes have been associated with numerous diseases and disorders. With modern biotechnology, these genes can be used as targets for the development of effective new therapies, which could significantly shorten the drug discovery process.
4. **Better vaccines:** Safer vaccines can be designed and produced by organisms transformed by means of genetic engineering. These vaccines will elicit the immune response without the attendant risks of infection.

Pharmaceutical Products

Most traditional pharmaceutical drugs are relatively simple molecules that have been found primarily through trial and error to treat the symptoms of a disease or illness. Biopharmaceuticals are large biological molecules such as proteins and these usually target the underlying mechanisms and pathways of a malady. They can deal with targets in humans that may not be accessible with traditional medicines.

Small molecules are manufactured by chemistry but larger molecules are created by living calls such as those found in the human body: for example, bacteria cells, yeast cells, animal or plant cells.

Modern biotechnology is often associated with the use of genetically altered microorganisms such as E. Coli or yeast for the production of substances like synthetic insulin or antibiotics. It can also refer to transgenic animals or transgenic plants, such as Bt corn.

Biotechnology is also commonly associated with landmark breakthroughs in new medical therapies to treat hepatitis B, hepatitis C, cancers, arthritis, haemophilia, bone fractures, multiple sclerosis and cardiovascular disorders. The biotechnology industry has also been instrumental in developing molecular diagnostic devices that can be used to define the target patient population for a given biopharmaceutical. Herceptin, for example, was the first drug approved for use with a matching diagnostic test and is used to treat breast cancer in women whose cancer cells express the protein HER2.

Modern biotechnology can be used to manufacture existing medicines relatively easily and cheaply. The first genetically engineered products were medicines designed to treat human diseases. To cite one example, in 1978 Genentech developed synthetic humanized insulin by joining its gene with a plasmid vector inserted into the bacterium Escherichia coli. Insulin, widely used for the treatment of diabetes, was previously extracted from the pancreas of abattoir (cattle and pigs). The resulting genetically engineered bacterium enabled the production of vast quantities of synthetic human insulin at a relatively low cost.

Genetic Testing

Genetic testing involves the direct examination of the DNA molecular for mutated sequences.

There are two major types of gene tests:

1. In the first type, a researcher may design short pieces of DNA (“probes”) whose sequences are complementary to the mutated sequences. These probes will seek their complement among the base pairs of an individual’s genome. If the mutated

sequence is present in the patient's genome, the probe will bind to it and flag the mutation.

2. In the second type, a researcher may conduct the gene test by comparing the sequence of DNA bases in patient's gene to disease in healthy individuals or their progeny.

Advantages of Genetic Testing

1. Carrier screening, or the identification of unaffected individuals who carry one copy of gene for a disease that requires two copies for the disease to manifest
2. Determining sex
3. Forensic/identify testing
4. New born screening
5. Prenatal diagnostic screening
6. Pre-symptomatic testing for estimating the risk of developing adult-onset cancers
7. Pre-symptomatic testing for predicting adult-onset disorders

Disadvantages of Genetic Testing

The absence of privacy and anti-discrimination legal protections in most countries can lead to discrimination in employment or insurance or other use of personal genetic information. This raises questions such as whether genetic privacy is different from medical privacy.

1. **Reproductive issues**: These include the use of genetic information in reproductive decision-making and the possibility of genetically altering reproductive cells that may be passed on to future generations. For example, germline therapy changes the genetic make-up of an individual's descendants. Thus, any error in technology or judgment may have far-reaching consequences. Ethical issues like designer babies and human cloning have also given rise to controversies between and among scientists and bioethicists, especially in light of past abuses with eugenics.

2. **Effects on social institutions:** Genetic tests reveal information about individuals and their families. Thus, test results can affect the dynamic within social institutions, particularly the family.
3. Conceptual and philosophical implications regarding human responsibility, free will vis-à-vis genetic determinism and the concepts of health and disease.

Recombinant DNA TECHNIQUE:

The recombinant DNA technological processes have had a great impact in the area of health care by enabling mass production of safe and more effective therapeutic drugs.

Further, the recombinant therapeutics do not induce unwanted immunological responses as is common is case of similar products isolated from non-human sources.

At present, more than 30 recombinant therapeutics have been approved for human-use the world over.

In India, around 12 of these are presently being marketed. Eg. (Genetically Engineered Insulin, Molecular Diagnosis, Disease control through GMO, Personal Genomics)

1. Genetically Engineered Insulin

Earlier, Insulin used for diabetes was extracted from the pancreas of slaughtered cattle and pigs.

1. It has caused patient of develop some kind of allergies or other kinds of reactions to the foreign protein.

Structure of Insulin

1. Insulin consists of two short polypeptide chains: Chain A and Chain B, that are linked together by disulphide bridges.
2. In Mammals, including humans, insulin – is synthesized as a prohormone (like a proenzyme, prohormone also needs to be processed before it becomes a fully mature and functional hormone) which contains an extra stretch called C peptide.
3. This C peptide is not present in the mature insulin and is removed during maturation into insulin.

4. The main challenge for production of insulin using rDNA technique was getting insulin assembled into a mature form.
5. How this was achieved through Biotechnology
6. In 1983, Eli Lilly, an American company, prepared two DNA sequences corresponding to A and B, chains of human insulin and introduced them in plasmids of E. Coli to produce insulin chains.
7. Chains A and B were produced separately, extracted and combined by creating bisulfide bonds to form human insulin.

2. Gene therapy/Human Genetic Engineering

Gene therapy is an experimental technique that uses genes to treat or prevent disease. In the future this technique may allow doctors to treat a disorder by inserting a gene into a patient's cells instead of using drugs or surgery. Researchers are testing several approaches to gene therapy, including

1. Replacing a mutated gene that causes disease with a healthy copy of the gene.
2. Inactivating, or "knocking out", a mutated gene that is functioning improperly.
3. Introducing a new gene into body to help fight disease.

Although gene therapy is a promising treatment option for some diseases (including inherited disorders, some types of cancer, and certain viral infections), the technique remains risky and is still under study. To make sure that it will be safe and effective. Gene therapy is currently being tested only for diseases that have no other cures.

Gene therapy is under study to determine whether it could be used to treat disease. Current research is evaluating the safety of gene therapy; future studies will test whether it is an effective treatment option. Several studies have already shown that this approach can have very serious health risks, such as toxicity, inflammation, and cancer.

Because the techniques are relatively new, some of the risks may be unpredictable; however, medical researchers, institutions, and regulatory agencies are working to ensure that gene therapy research is as safe as possible. Because gene therapy involves making changes to the body's set of basic instructions, it raises many unique ethical concerns.

Current gene therapy research has focused on treating individuals by targeting the therapy to body cells such as bone marrow or bloods cells. This type of gene therapy cannot be passed to a person's children. Gene therapy could be targeted to egg and sperm cells (germ cells), however, which would allow the inserted gene to be passed to future generations. This approach is known as germline gene therapy.

The idea of germline gene therapy is controversial. While it could spare future generations in a family from having a particular genetic disorder, it might affect the development of a fetus in unexpected ways or have long-term side effects that are not yet known. Because people who would be affected by germline gene therapy are not yet born, they can't choose whether to have the treatment.

Its application is huge. They're trying to grow transplantable human organs inside pigs. They're even hatching mutant mosquitoes designed to be incapable of spreading malaria and exploring ways to wipe out invasive species. Scientists are engineering animals with humanlike disorders to unravel the gene defects that fuel them. They're building stronger immune cells, and developing potential treatments for muscular dystrophy, sickle cell disease and cancer.

Recent Developments

1. **The United States Approved first Gene Therapy in the Nation** (Sep 2017): - US Food and Drug Administration approved gene therapy for Children and Young Adult patients up to the age of 25 years with a form of acute lymphoblastic leukaemia, a kind of cancer. This treatment will use a patient's own immune cell, called T- Cells, along with white blood cells. These cells are removed from a patient, sent to a lab, and encoded with a viral vector, reprogrammed, and returned to the patient. Studies have shown that 83% of the patient respondent to the treatment, achieving remission within three months.

   The treatment is made by Novartis and is called Kymriah (tisagenlecleucel)

   Significance – It is a historic action which will usher in a new approach to the treatment of cancer and other serious and life-threatening diseases.

2. **Gene editing:** Gene editing or genome editing is a way of making specific changes to the DNA of a cell or organism. An enzyme cuts the DNA at a specific sequence, and when this is repaired by the cell, a change or 'edit' is made to the sequence. Enzymes which cut DNA are known as engineered nucleases. DNA is inserted, deleted or replaced in the genome of a living organism using engineered nucleases or molecular scissors. Currently, there are four families of engineered nucleases which as being used

1. Mega nucleases,
2. Zinc finger nucleases (ZFNs),
3. Transcription activator-like effector-based nucleases (TALEN).
4. Clustered regularly interspaced short palindromic repeats (CRISPR)-Cas system.

These nucleases create site-specific double-strand breaks (DSBs) at desired locations in the genome.

CRISPR-Cas9

CRISPR is a dynamic, versatile tool that allow us to target nearly any genomic location and potentially repair broken genes. It can remove, add or alter specific DNA sequences in the genome of higher organisms.

CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) are sections of DNA and are sections of genetic code containing short repetitions of base sequences followed by spacer DNA segments.

CAS-9 (CRISPR-associated protein 9) is an enzyme. It uses a synthetic guide RNA to introduce a double-strand break at a specific location within a strand of DNA. It is a system used by bacterial cells to recognize and destroy viral DNA as a form of adaptive immunity.

CRISPR-Cas9 works

CRISPR scans the genome looking for the right location and then uses the Cas9 protein as molecular scissors to snip through the DNA.

Cas9 endonuclease – guide RNAs to direct it to a particular sequence to be edited. The genetic sequence of the RNA matches the target sequence of the DNA that has to be edited.

When Cas9 cuts the target sequence, the cell repairs the damage by replacing the original sequence with an altered version.

Unlike other gene-editing methods, it is cheap, quick, easy, safer and more accurate to use because it relies on RNA-DNA base pairing, rather than the engineering of proteins that bind particular DNA sequences.

Pros of gene editing

CRISPR could be used to modify disease-causing genes in embryos brought to term, removing the faulty script from the genetic code of the person's future descendants as well. Gerome editing (Gene editing) could potentially decrease, or even eliminate, the incidence of many serious genetic diseases, reducing human suffering worldwide. It might also be possible to install genes that offer lifelong protection against infection.

Cons of Gene editing

Making irreversible change to every cell in the bodies of future children and all their descendants would constitute extraordinarily risky human experimentation.

There are issues including off-target mutations (unintentional edits to the genome), persistent editing effects, genetic mechanisms in embryonic and fetal development, and longer-term health and safety consequences.

Some argue that we do not understand the operations of the genome enough to make long-lasting changes to it. Altering one gene could have unforeseen and widespread effects on other parts of the genome, which would then be passed down to future generations.

Many consider genome alterations to be unethical, advocating that we should let nature run its course.

Few argue that after permitting human germline gene editing for any reason would likely lead to its ignorance of the regulatory limits, to the emergence of market-based eugenics that would exacerbate already existing discrimination, inequality, and conflict.

It will become a tool for selecting desired characteristics such as intelligence and attractiveness.

Risks associated with Gene Editing

In the 2016 Worldwide Threat Assessment of the US Intelligence Community Statement United States Director of National Intelligence, James R. Clapper, named genome editing as a potential weapon of mass destruction, stating that genome editing conducted by countries with regulatory or ethical standards different from Western countries probably increases the risk of the creation of harmful biological agents or products.

Low cost and accelerated pace of development of this technology and its deliberate or unintentional misuse might lead to far-reaching economic and national security implications.

It could lead to the manufacture of biological weapons by potential bioterrorists who might use the knowledge to create vaccine-resistant strains of other pox viruses, such as smallpox, that could affect humans.

3. Molecular Diagnosis:

2. For treatment of any disease, early diagnosis and understanding its pathophysiology is very important. Using conventional methods of diagnosis (serum and urine analysis, etc.) early detection is not possible.
3. Recombinant DNA technology, Polymerase Chain Reaction (PCR) and Enzyme-linked immune-Sorbent Assay (ELISA) are some of the techniques that serve the purpose of early detection.
    i. PCR is a technique used in molecular biology to amplify a single copy or a few copies of a piece of DNA across orders of magnitude, generating thousands to millions of copies of a particular DNA sequence. It is now

routinely used to detect HIV in suspected AIDS patients. It is being used to detect mutations of genes in suspected cancer patients tog.

ii. ELISA is based on the principle of antigen-antibody interaction. Infection by the pathogen can be detected by the presence of antigens (Proteins, glycoproteins etc.) or by detecting the antibodies synthesized against the pathogens.

4. Disease Control Through Genetically Modified Organisms

1. By introducing sterile mosquitoes (genetically formed), (concept – not done yet)
2. Synthetic vector genome which is incapable of hosting the parasite and virus.

TRANSGENIC ORGANISMS

A genetically modified organism (GMO), is an organism whose genetic material has been altered using genetic engineering techniques (i.e., a genetically engineered organism). GMOs are used to produce many medications and genetically modified foods and are widely used in scientific research and the production of other goods. The term GMO is very close to the technical legal term, 'living modified organism' defined in the **Cartagena Protocol on Biosafety.**

A more specifically defined type of GMO is a 'transgenic organism". This is an organism whose genetic makeup has been altered by the addition of genetic material from an **unrelated organism.** Typically, GMOs are organisms whose genetic makeup has been altered without the addition of genetic material from an unrelated organism.

Cartagena Protocol

The Cartagena Protocol of Biosafety to the Convention on Biological Diversity is an international agreement on biosafety Protocol seeks to protect biological diversity from the potential risks posed by genetically modified organisms resulting from modern biotechnology. It basically regulates international trade in living GMOs (specially, 'any living organism that possesses a novel combination of genetic material obtained through the use of modern biotechnology).

Animals: Genetically modified animals are being developed for the following purposes.

1. To research human diseases (for example, to develop animal models for these diseases).
2. To produce industrial or consumer products (fibres for multiple uses);
3. To produce products intended for human therapeutic use (pharmaceutical products or tissue for implantation);
4. To enrich or enhance the animals' interactions with humans (hypo-allergenic pets);
5. To enhance production or food quality trails (faster-growing fish, pigs that digest food more efficiently);
6. To improve animal health (disease resistance)

Uses: Transgenic cows for higher milk yield can be developed by activating the casein the milk protein gene. Transgenic sheep for higher wool yield can be developed by activating the keratin the hair protein gene. Transgenic fish that grows larger in size can be developed by activating the all fish growth regulator gene.

Prelims fact: The first genetically modified animal to be commercialized was the GloFish, a Zebra fish with a fluorescent gene added that allows it to glow in the dark under ultraviolet light. The first genetically modified animal to be approved for food use was AquAdvantage salmon in 2015. The salmon was transformed with a growth hormone-regulating gene from a Pacific Chinook salmon and a promoter from an ocean pout enabling it to grow year-round instead of only during spring and summer.

Bacteria:

GM Bacteria were the first organisms to be genetically modified in the laboratory, due to the relative ease of modifying their chromosomes. Genetically modified bacteria are used to produce large amounts of proteins for industrial use. The first medicinal use of GM bacteria was to produce the protein insulin to treat diabetes. They have been used to produce biofuels. Other uses for genetically modified bacteria include bioremediation, where the bacteria are used to convert pollutants into a less toxic form E.g., Oil Zapper developed by The Energy and resources Institute (TERI). It consists of five different bacterial strains. Oil zapper feeds on hydrocarbon compounds present in crude of and the hazardous hydrocarbon waste generated by oil refineries, known as Oil Sludge and the mixture into harmless CO_2 and water. Research is on whether

engineered bacteria can treat conditions that affect the brain, liver and other organs – and even kill other, harmful microbes.

Genetically Engineering Approval Committee (GEAC)

Genetic engineering appraisal committee (GEAC) is Apex body under environment ministry for regulating, manufacturing, use, import, export anti storage of hazardous microorganisms are genetically engineered organisms and cells in the country. The committee functions as a statutory body under the Environment Protection Act 1986 of the Ministry of Environment & Forests (MoEF).

Its functions are:

1. It is responsible for giving technical approvals of the proposal relating to the release of GMOs and products including experimental field trials.
2. Under the EPA 1986 'Rules for Manufacture, Use, Import, Export and Storage of Hazardous Microorganisms Genetically Engineered Organisms or Cells 1989', GSAC is responsible for granting permits to conduct experimental and large-scale open field trials and also approve commercial release of biotech crops.
3. The committee also had the power to take punitive action under the Environment Protection Act.

Issues with GEAC:

1. GEAC is a body made of bureaucrats having no expert knowledge of highly complex issues that arise in biotechnology. Thus, this mode of regulation is considered faulty and insufficient.
2. No provision of taking into consideration the option of masses and those affected by GEAC's decisions. Also, no provision of civil society participation and thus wasn't people-oriented.
3. There is no mechanism for an appeal. No adjudicating authority is present for biotechnology specific issues.
4. Though agriculture is a state subject, no authority is given to the state governments.

5. Procedure for approval of a particular modified crop is not stated which makes the whole process vague and questionable.

To overcome these lacunas and to have a sound regulatory mechanism for biotechnology in India, the government has introduced the Biotechnology Regulatory Authority of India Bill 2013. This has been in limbo since as the bill has met with staunch opposition.

ADVANCEMENTS IN HORTICULTURE & AGRICULTURE:

Introduction:

Using the techniques of modern biotechnology, one or two genes may be transferred to a highly developed crop variety to impart a new character that would increase its yield.

The Characters are

1. Reduced Vulnerability of Crops to Environmental Stresses
2. Increased Nutritional Qualities
3. Improved Taste, Texture or Appearance of Food
4. Reduced Dependence on Fertilizers, and other Agrochemicals
5. Production of Novel Substances

Reduced Vulnerability of Crops to Environmental Stresses

Crops containing genes that will enable them to withstand biotic and abiotic stresses may be developed. For example, drought and excessively salty soil are two important limiting factors in crop productivity. Biotechnologists are studying plants that can cope with these extreme conditions in the hope of finding the genes that enable them to do so and eventually transferring these genes to the more desirable crops.

Increased Nutritional Qualities

Proteins in foods may be modified to increase their nutritional quantities. Proteins in legumes and cereals may be transformed to provide the amino acids needed by human being for a balanced diet. For example, Golden rice.

Improved Taste, Texture or Appearance of Food

Modern biotechnology can be used to slow down the process of spoilage so that fruit can ripen longer on the plant and then be transported to the consumer with a still reasonable shelf life. This alters the taste, texture and appearance of the fruit. It could expand the market for farmers in developing countries due to the reduction in spoilage. For example, engineering soybeans to resist spoilage makes them less suitable for producing tempeh which is a significant source of protein that depends on fermentation.

Reduced Dependence on Fertilizers, and other Agrochemicals

Most of the current commercial applications of modern biotechnology in agriculture are reducing the dependence of farmers on agrochemicals. For example, Bacillus thuringiensis (Bt) is a soil bacterium that produces a protein with insecticidal qualities.

The lack of herbicides with broad – spectrum activity and no crop injury were a consistent limitation in crop ween management. The introduction of herbicide-tolerant crops has the potential of reducing the number of active herbicide ingredients used for weed management, reducing the number of herbicide applications made during a season and increasing yield due to improved weed management and less crop injury.

Production of Novel Substances:

In Crop Plants Biotechnology is being applied for novel uses other than food. For example, oilseed can be modified to produce fatty acids for detergents, substitute fuels and petrochemicals. Potatoes, tomatoes, rice tobacco, lettuce, safflowers, and other plants have been genetically engineered to produce insulin and certain vaccines.

GM CROPS

Genetically modified crops (GM crops, or biotech crops) are plants used in agriculture, the DNA of which has been modified using genetic engineering techniques. The aim is to introduce a new trial to the plant which does not occur naturally in the species.

Though they are a success worldwide, in India, GM crops have not been accepted and approved.

The only genetically modified cash crop under commercial cultivation in India is cotton. GM crops were first commercialized in the US, Canada, Mexico, Argentina, China and Australia in 1996, and in 2016 more than 1.8 crore farmers in 26 countries planted GM crops. Four-fifths of the world's soybean crop are GM, as are two-thirds of cotton and a third of maize; India, the world's biggest cotton producer, has the fifth largest area under GM crop cultivation, and BT cotton seeds account for 40% of the Rs. 14,000 crore national seeds market. Maharashtra's neighbor Gujarat grows more cotton than any other state.

Necessity: GMOs benefit not only farmers and the public, but also biodiversity, soil quality, water quality, carbon sequestration – in short, the environment. High yields from GMOs curtail the conversion of forest of grassland to agriculture. Benefits of GM Crops are many. Genetic engineering can improve crop protection. Crops with better resistance to pest and diseases can be created. The use of herbicides and pesticides can be reduced or even eliminated. Farmers can achieve high yield and thereby get more income.

Nutritional content can be improved. The shelf life of foods can be extended. Food with better taste and texture can be achieved. Crops can be engineered to withstand extreme weather. Climate change is likely to make the problem far worse, bringing higher temperatures and, in many regions, wetter conditions that spread infestation of disease and insects into new areas. Drought, damaging storms, and very hot days are already taking a toll on crop yields, and the frequency of these events is expected to increase sharply as the climate warms. For farmers, the effects of climate change can be simply put: the weather has become far more unpredictable, and extreme weather has become far more common. Thus, we can say that the rapidly growing global population and a warming climate will make agricultural innovations a necessity.

Unintended side effects: Genetic engineering is a new field, and long-term results are unclear. Very little testing has been done on GM food. Some crops have been engineered to create their toxins against pests. This may harm non-targets such as farm

animals that ingest them. The toxins can also cause allergy and affect digestion in humans.

Further, GM crops are modified to include antibiotics to kill gems and pests. And when we eat them, these antibiotic markers will persist in our body and will render actual antibiotic medications less effective over a period of time, leading to superbug threats. This means illnesses will become more difficult to cure.

Besides, health and environmental concerns, there is serious concern about multinational agribusiness companies taking over farming from the hands of small farmers. Dependence on GM seed companies could prove to be a financial burden for farmers. Farmers are reluctant because they will have limited rights to retain and reuse seeds as the multinational agribusiness companies use Terminator seeds. Their concern also includes finding a market that would accept GM food as the general public is very of using genetically modified crops.

The multinational agribusiness companies use techniques to enhance their business. Some of them are:

Use of Terminator seeds- Terminator Technology allows a modification of plants to make them produce sterile seeds thus making them one time use only forcing the farmer to buy them again and again.

Use of traitor Technology – this technology requires application of chemical to the GM crop to reactivate engineer to trade such as flowering, herbicide, resistance etc.

BT Cottons

BT cotton was first used in India in 2002. Cotton yield more than doubled in the first decades since its introduction in 2002. But the success of the crop in the Indian context is debatable as the modified cotton too, prone to pest attacks. E.g. Pink Ball worm attack on BT cotton in Maharashtra in 2017. Monsanto introduced its first-generation BT cotton, called Bollgard I (BG-I) in 2002 and Bollgard it (BG-II) in 2006, the latter of which is still the de facto GM cotton variety.

Traditional hybrid seeds are a result of cross-pollination of two different but related plants, but GM hybrid seeds involve the artificial insertion of a gene from a different species. For instance, BG-II contains two genes from the bacterium Bacillus thuringiensis (hence BT cotton), and BG-I contains on the only gene.

They are supposed to provide immunity to the plant from pests like American and pink bollworms GM seeds are mostly either pest-resistant or herbicide-tolerant, the latter of which is still not legal in India. Despite the wide adoption of BT cotton, India has been wary of GM food crops, withholding its nod for brinjal and mustard.

The pink bollworm is a manageable insect and that it be can be tackled by opting for shorter duration crops – 140-160 days. But the issue in India is that the life of the crop is usually more than 180 days to grow them longer and thus yield more cotton.

Last year, Division Bench of the Delhi High Court had invalidated the patent granted to Monsanto-acquired by German firm Bayer AG – saying the gene sequence responsible for the Bt trait that eradicates pests afflicting cotton plants is a part of the seed, and hence, cannot be patented under Section 3(i) of the Patents Act, 1970. This ruling was overturned by the Supreme Court recently. The Supreme Court allowed Monsanto Technology to claim a patent on its genetically-modified cotton seeds, giving a boost to firm developing new seed technologies.

BT Brinjal:

Bt brinjal contains a 'cry1Ac' gene isolated from the soil bacterium Bacillus thuringiensis. The gene codes for a protein toxic to the fruit and shoot borer (FSB), a major insect pest that attacks the brinjal crop throughout its life cycle. The technology for Bt brinjal- the process of integrating the foreign gene into the host plant, which includes developing the protocols as well as identifying the specific location in the genome where the insertion is to take place – was developed by the Jalna-based Maharashtra Hybrid Seeds Company (Mahyco).

The Genetic Engineering Appraisal Committee (GEAC) in 2007 recommended the commercial release of Bt brinjal. But the initiative was blocked in 2010 because Bt brinjal poses risks to the environment and possibly to human health. The occurrence of wild, weedy and also cultivated relatives presents a likelihood that the GE Bt gene will

spread to these relatives. The spread of the GE Bt gene would have considerable ecological implications, as well as implications for future crop contamination and farmers' rights. Also, the spread of the GE Bt gene could result in the brinjal becoming an aggressive and problematic weed.

Recently in 2013, BARI (Bangladesh Agriculture Research Institute) has approval the use of GM brinjal. This has been a cause of concern for India because India and Bangladesh share a porous border and thus there is a risk of the BT gene spreading to the Indian brinjal crops.

GM Mustard

GM Mustard, aka DMH-11 or Dhara-11, is developed by Delhi University. It is a genetically modified variety of Herbicide Tolerant (HT) mustard. GM mustard is being developed to increase the productivity of mustard to meet the ever-growing demand for edible oil in the country. IN a submission to the Environment Ministry, the Genetic Engineering Appraisal Committee has given a positive recommendation but with certain conditions.

With the GEAC nod, the GM mustard developed by the Delhi University gets closer to become India's first edible – GM crop. 'Several groups are opposing the GEAC's decision as it might affect allied agricultural activities, the health of the people etc.

The Centre for Genetic Manipulation of Crop Plants (CGMCP), Delhi University South Campus, had submitted an application to the GEAC for the environmental release of GM mustard (Brassica Juncea), hybrid DMH-11 and the use of parental events (varunabn 3.6 and EH2 mod bs 2.99) for the development of a new generation of hybrids.

Pros:

1. A system of genes from a soil bacterium that makes mustard have been used by the scientists. This is generally a self-pollinating plant and is better suited to hybridization than current methods.
2. Local crop developers can more easily develop different varieties of hybrid mustard, like GM cotton, and confer traits like pest resistance and potentially

improving yield. India's dependence on edible oil imports makes it necessary to harness GM-mustard. About $12 billion is spend by India annually on the import of edible oil.

3. As the population will increase, India will have to increase its production of oilseed crops.
4. DMH-11 is claimed to be one of the promising technologies to improve mustard yield in India, which has been almost stagnant since the last two decades.

Cons:

1. India needs technological interventions in agriculture and other fields, but the issue of genetically modified crops has always been controversial. Mustard is used for edible oil, their leaves are eaten, and it is also used as ayurvedic medicine. Its impact on the health of the people, environment, soil, groundwater or food chain is not known yet. The glufosinate-based herbicide to which the proposed GM mustard is tolerant will also have adverse impacts on health. An herbicide-tolerant crop promotes constant exposure to a single herbicide which eventually results in weeds becoming resistant.
2. There are social and economic concerns attached to any technology as well. In this case, farmers, especially in Punjab and other mustard growing areas, have said that they don't need new varieties. They need better policy, pricing and to rationalize the input costs against the cost which they get after selling it in the markets. If these problems cannot be solved, bringing in another technology might not solve the problem.
3. GEAC needs to be a transparent body. It should put it in the public domain that on what grounds it has approved GM mustard for citizens to analyse and stay aware whether it is good for them or not. If there is a lack of trust on the part of people who are to use it, it will be like forcing something upon them. India is a signatory to Cartagena Protocol on biosafety where it has committed to public participation in decision making.
4. There have to be strong liability laws. If there are any environmental hazards or if something goes wrong in future which is not there in India at present. The

pesticide industry's efforts to influence policy makers & regulators have obstructed reforms globally. Their business model aims only at making a profit.

5. Since almost 70 per cent of the mustard crop is cultivated in Rajasthan, Madhya Pradesh and Haryana, the problem farmers face is that of over-production and lack of buyers. Increasing production of oilseed crops like mustard, therefore, requires adequate policy initiatives. Raising import tariffs to ensure that cheaper edible oil does not flood the market, and providing an economic price for the produce can bring back self-sufficiency in dibble oils.
6. Allowing the cultivation of GM mustard would lead to a direct attack on women involved in the mustard crop weeding.

FSSAI DRAFT FOOD SAFETY AND STANDARDS (LABELLING AND DISPLAY) REGULATION 2018

In India for the first time seeking to introduce GMO labelling, the country's food regulator has proposed all packaged food products containing genetically modified (GM) ingredients must clearly state it on their labels. The Food Safety and Standards Authority of India (FSSAI) also suggests mandatory declaration by packaged food manufacturers about nutritional information such as calories, total fat, trans fat, sugar and salt per serve on the front of the pack.

Presently the country has no provision for GM labeling in its regulatory mechanism. However, FSSAI had released draft notice – Food Safety and Standards (Labelling and Display) Regulations, 2018 making it mandatory to label such foodstuffs as 'Contains GMO/Ingredients derived from GMO" if such items contain 5% or more GE ingredients.

Conclusion:

Agriculture is a state subject. Therefore, it is important for the Centre to late into consideration the views of State Governments as well, it is expected that the Supreme Court will ensure the protection of Indian consumers and farmers as giving a nod to GM mustard will pave the way for clearance to other GM crops as well whether the impact is good or bad.

Questions:

1. Give a detailed account on gene editing.
2. Explain about the
   a) Genetically modified organism.
   b) Genetically modified bacteria.
3. Describe in detail about the Genetically modified crops.